



Smart Academic Attendance and Automated Compliance Reporting System

Diploma Version

Swinburne University of Technology Sarawak

Application Team

Problem Statement

While Qwickly Attendance is deployed in higher education institutions as a backend attendance engine, relying solely on generic third-party apps creates fragmentation in the student and staff user experience. Off-the-shelf Qwickly applications function as standalone tools and cannot be branded, customized, or natively embedded into a centralized **Swinburne Student Portal**.

Furthermore, off-the-shelf mobile solutions do not natively bridge Qwickly attendance records with university Student Information System (SIS) APIs. Generating regulatory reports—such as mandatory attendance compliance monitoring for international students on student visas—requires extracting raw attendance logs from Qwickly and manually merging them with student visa metadata. This process is time-consuming, prone to error, and disconnected from Swinburne's internal administrative workflows.

Therefore, there is a need for a dedicated, Swinburne-branded React Native mobile application that connects directly to the **Qwickly REST API**, providing a unified attendance experience within the Swinburne ecosystem, supported by an **n8n automation engine** to streamline compliance PDF generation and class Excel exporting.

Project Description

This project aims to develop a custom Swinburne-branded mobile app integrated with Qwickly's API infrastructure and an automated n8n reporting engine designed for a 12-week development timeframe.

Phase 1: Custom Swinburne React Native Mobile App

Phase 1 focuses on building a cross-platform mobile application (using React Native / Expo) that serves as part of the unified Swinburne Student Portal ecosystem, using Qwickly as the underlying attendance backend via its REST API:

- **Swinburne SSO & Role Authentication:** Authenticates users via university credentials, routing students and lecturers to their respective portals.
- **Lecturer Interface:** Lecturers can launch class sessions, display dynamic QR codes, and view real-time class lists. Instructors can manually edit or override student attendance statuses (Present, Late, Absent, Excused) by sending updates directly to Qwickly's endpoints.

- **Student Interface:** Students can scan session QR codes directly from the Swinburne App to mark themselves present in Qwickly, while also viewing their individual attendance history and percentages across all enrolled units.

Phase 2: n8n Automated Compliance Workflows

Phase 2 implements two automated n8n workflows triggered via webhooks to synthesize Qwickly data with central student database records:

1. **Workflow 1: International Student Visa Compliance PDF** — Triggered to query attendance logs from Qwickly, merge them with student visa details from the Student Data API via sis_id, and generate an official audit-ready PDF compliance report.
2. **Workflow 2: Class Attendance Excel Export** — Triggered to pull full course attendance logs from Qwickly and format them into a structured Excel (.xlsx) workbook for departmental archiving.

Please note that a comprehensive requirement gathering process should be conducted to validate and finalize all project specifications.

Features

Feature	Description
Role-based Access Control	Login using Microsoft.
Cross-Platform Mobile App	React Native (Expo) app for iOS and Android delivering mobile attendance check-in.
Notification and Alert feature	Alert student for announcement and alerts.
QR Code Attendance Record	Camera scanner allowing students to post attendance check-ins directly to the Qwickly REST API.
Student Attendance Dashboard	Displays individual attendance summary metrics retrieved from Qwickly course endpoints.
Lecturer Real-Time Override	Interface for instructors to display session QR codes and update Qwickly attendance records manually.
n8n Workflow 1: Visa Compliance PDF	Automated workflow merging Qwickly attendance with Student Data API visa statuses to generate official PDF reports.
n8n Workflow 2: Class Excel Export	Automated workflow querying Qwickly course logs to convert JSON payloads into downloadable .xlsx files.